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(54) **METHOD FOR MANUFACTURING WET FRICTION PLATE**

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B23K 26/364 (2014.01)
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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2002/0127379 A1* 9/2002 Suzuki H05K 1/0366 428/209
2007/0270069 A1* 11/2007 Lee F16D 69/02 66/190
(Continued)

FOREIGN PATENT DOCUMENTS

JP 11-247904 A 9/1999
JP 2010-48272 A 3/2010
(Continued)

OTHER PUBLICATIONS

International Search Report (ISR) dated mailed Jul. 13, 2021 filed in PCT/JP2021/016346.

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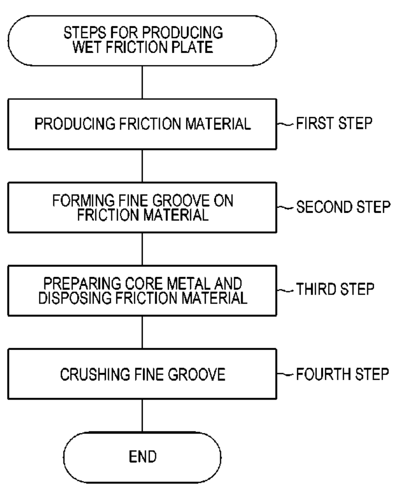
(57) **ABSTRACT**

There is provided a method for producing a wet friction plate that can improve the retention property of lubricating oils in a fine groove formed by laser light. In a method for producing a wet friction plate (**200**), a friction material (**210**) is produced by sheet making processing in a first step. Thereafter, in a second step, a fine groove (**211**) is formed on the friction material (**210**). The friction material (**210**) is produced such that thermosetting resin therein is in a semi-cured state. The cross-sectional shape of the fine groove (**211**) is formed into a V shape by laser light. Subsequently, in a third step, the friction material (**210**) is disposed on a core metal (**201**) via an adhesive agent including thermo-setting resin. Subsequently, in a fourth step, the friction material (**210**) is pressed while heated to crush the fine groove (**211**). In this manner, unevenness is formed on an
(Continued)



intra-groove surface (**212**), and the thermosetting resin is completely cured.

7 Claims, 6 Drawing Sheets

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F16D 13/72	(2006.01)
F16D 13/74	(2006.01)
F16D 69/00	(2006.01)
F16D 69/02	(2006.01)

(52) **U.S. Cl.**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

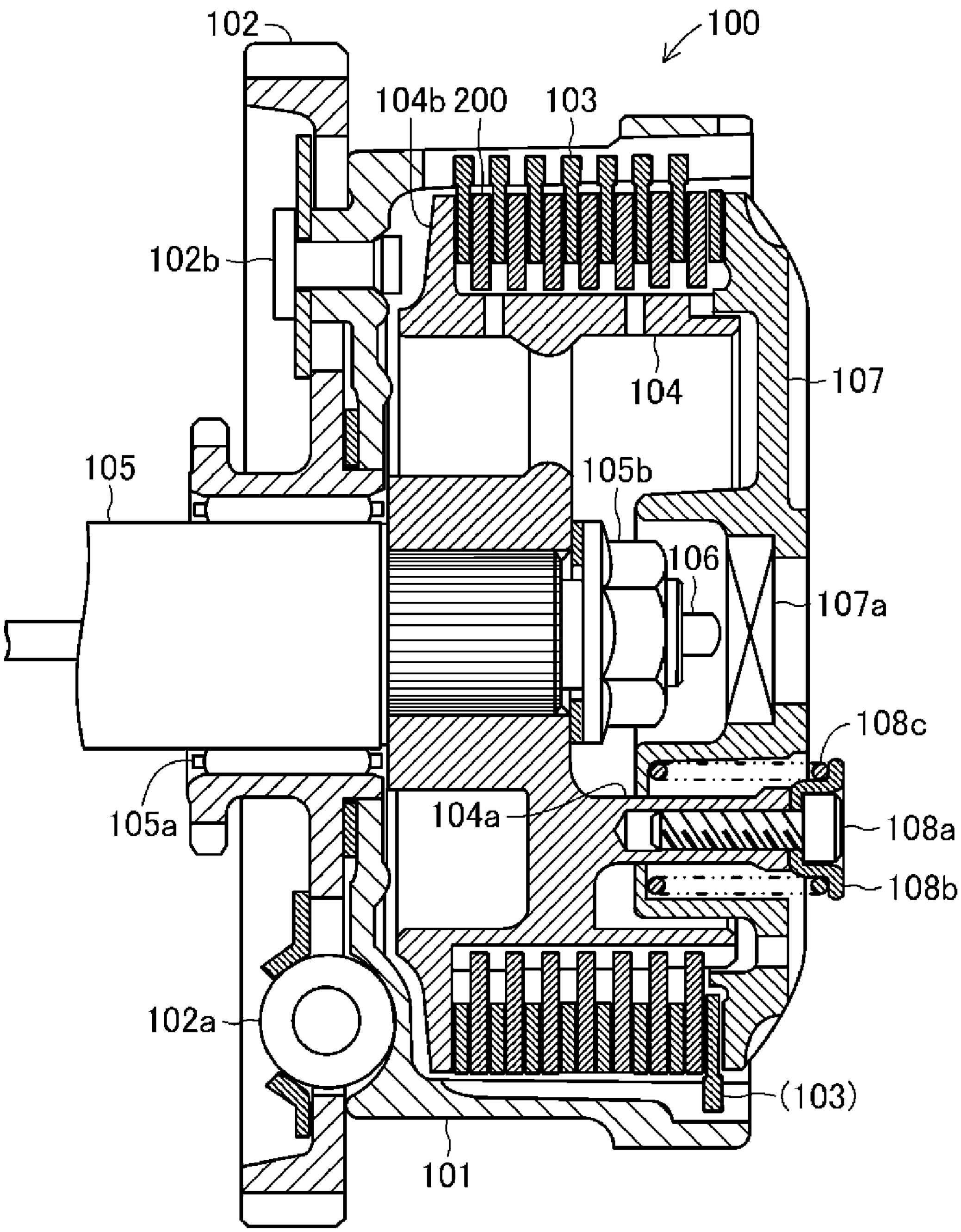
2010/0044182	A1	2/2010	Sakabe et al.	
2013/0168199	A1	7/2013	Higashijima et al.	
2019/0193207	A1*	6/2019	Kremer	F16D 65/127

FOREIGN PATENT DOCUMENTS

JP	2014-133242	A	7/2014
WO	2011/118347	A1	9/2011

* cited by examiner

FIG. 1



100
102
104b
200
103
102b
104
107
105
105b
106
107a
108c
105a
104a
108a
108b
102a
(103)
101

FIG. 2

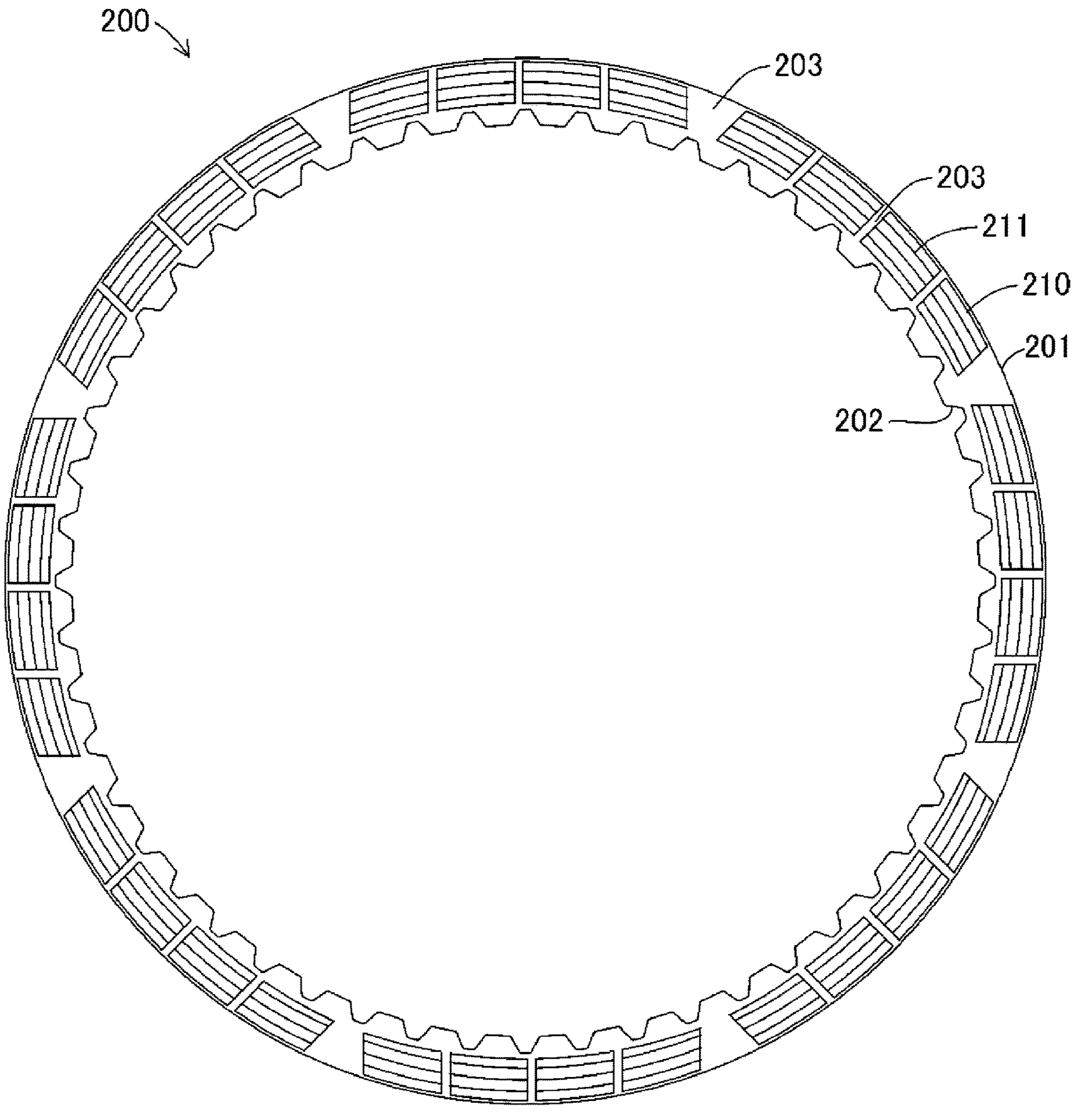


FIG. 3

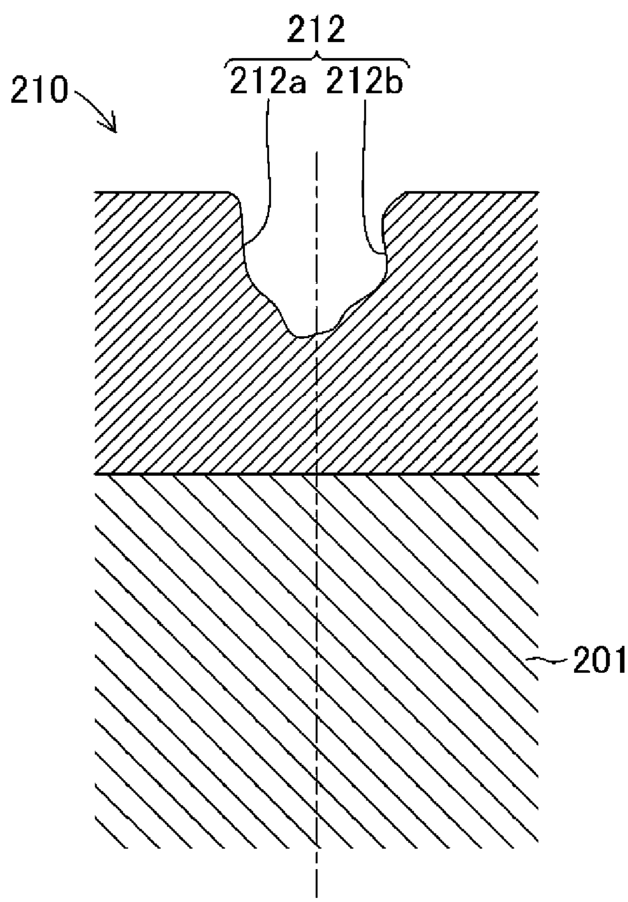


FIG. 4

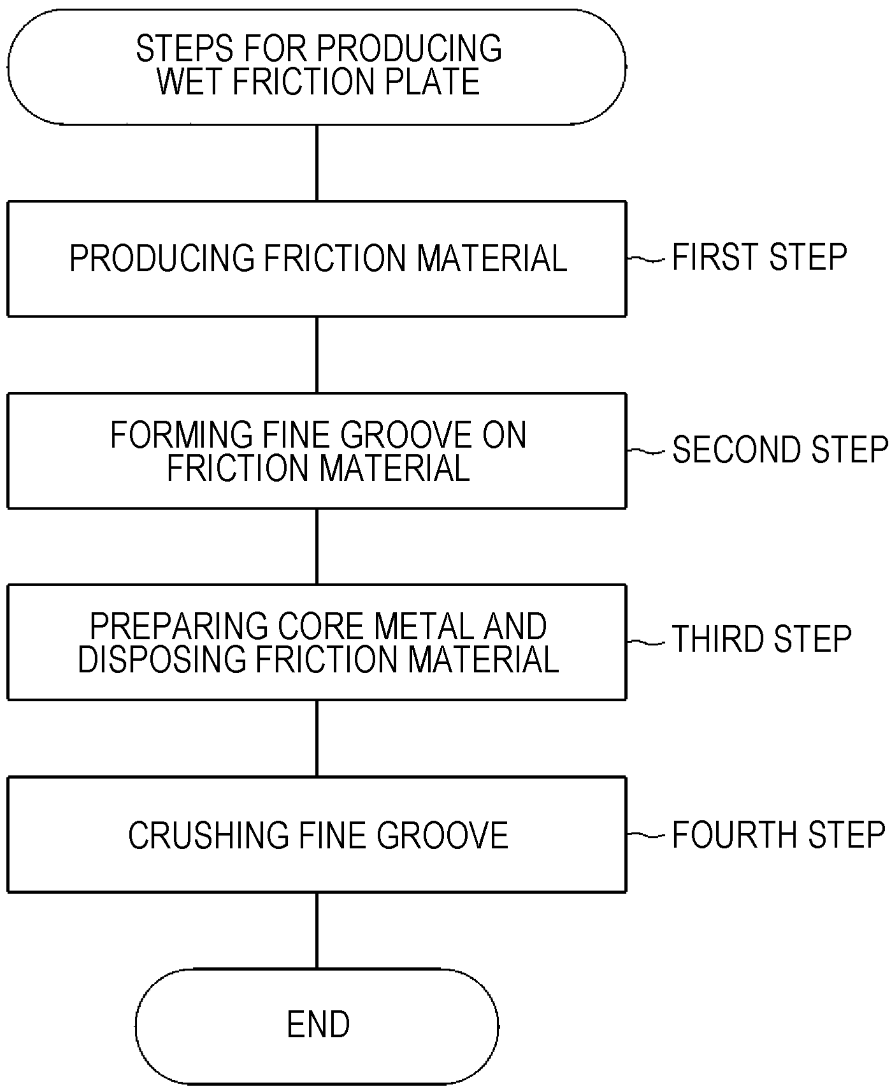

STEPS FOR PRODUCING WET FRICTION PLATE
PRODUCING FRICTION MATERIAL
FIRST STEP
FORMING FINE GROOVE ON FRICTION MATERIAL
SECOND STEP
PREPARING CORE METAL AND DISPOSING FRICTION MATERIAL
THIRD STEP
CRUSHING FINE GROOVE
FOURTH STEP
END

FIG. 5

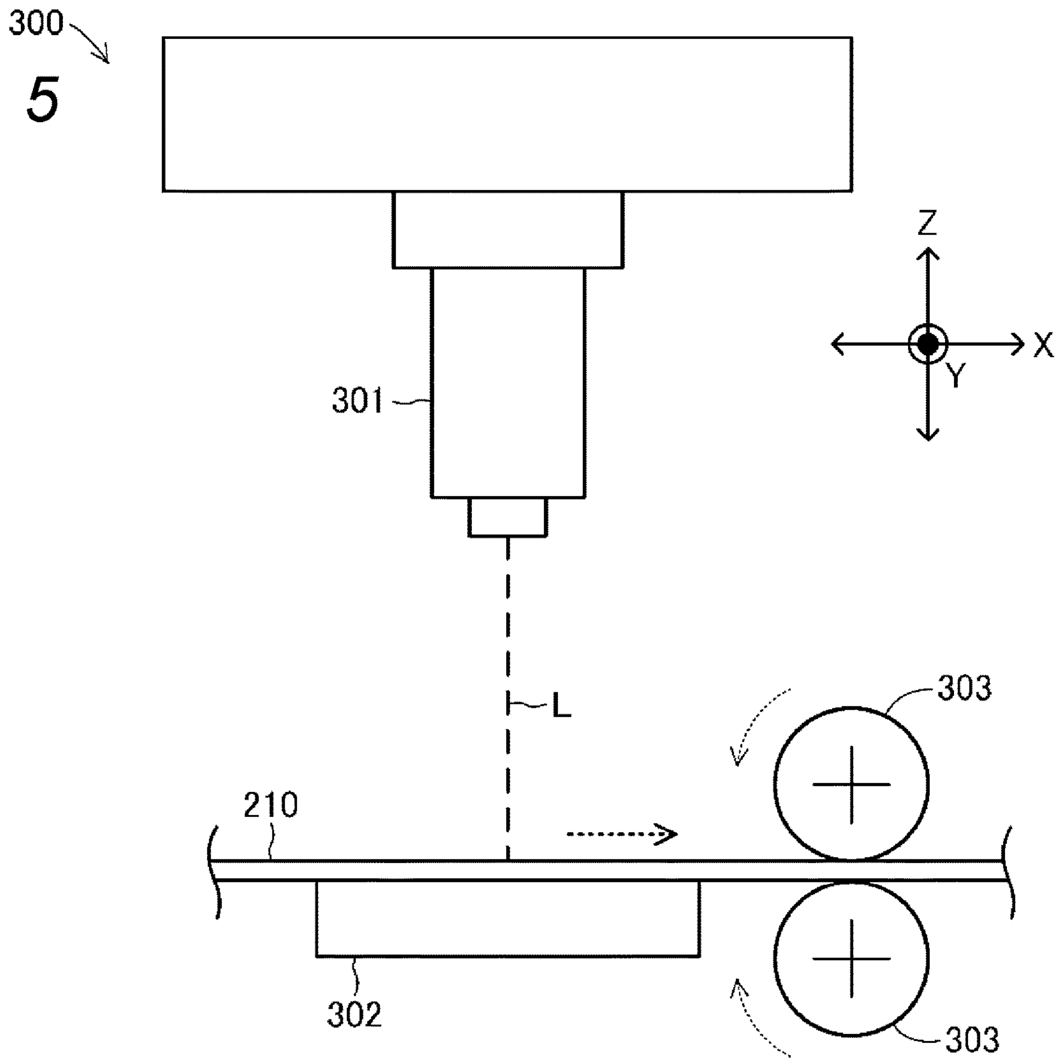


FIG. 6

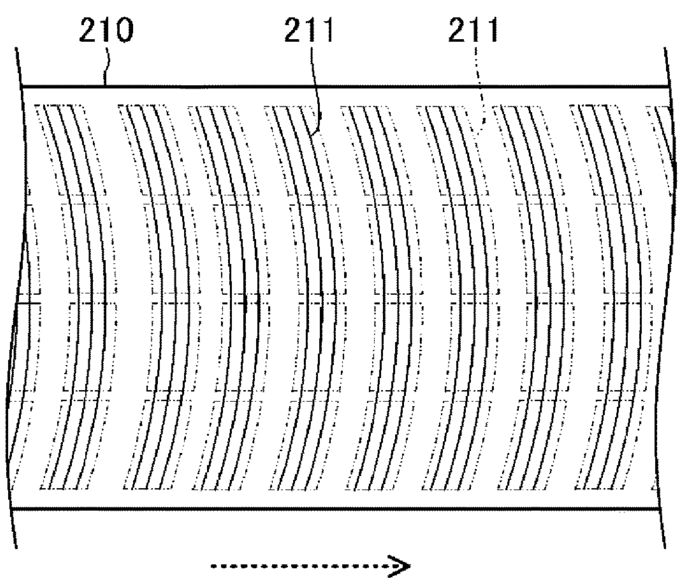


FIG. 7

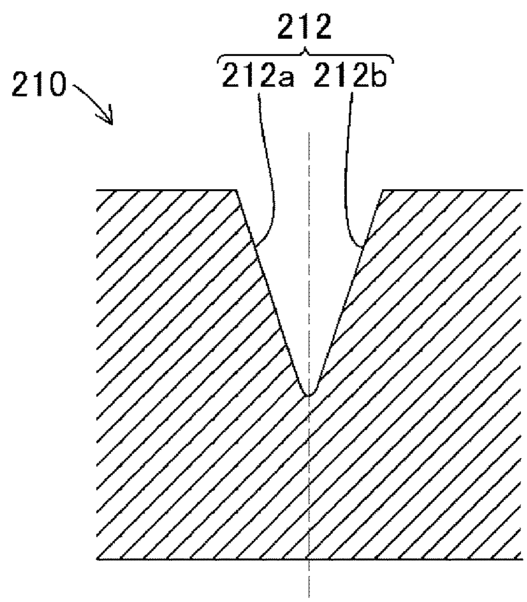
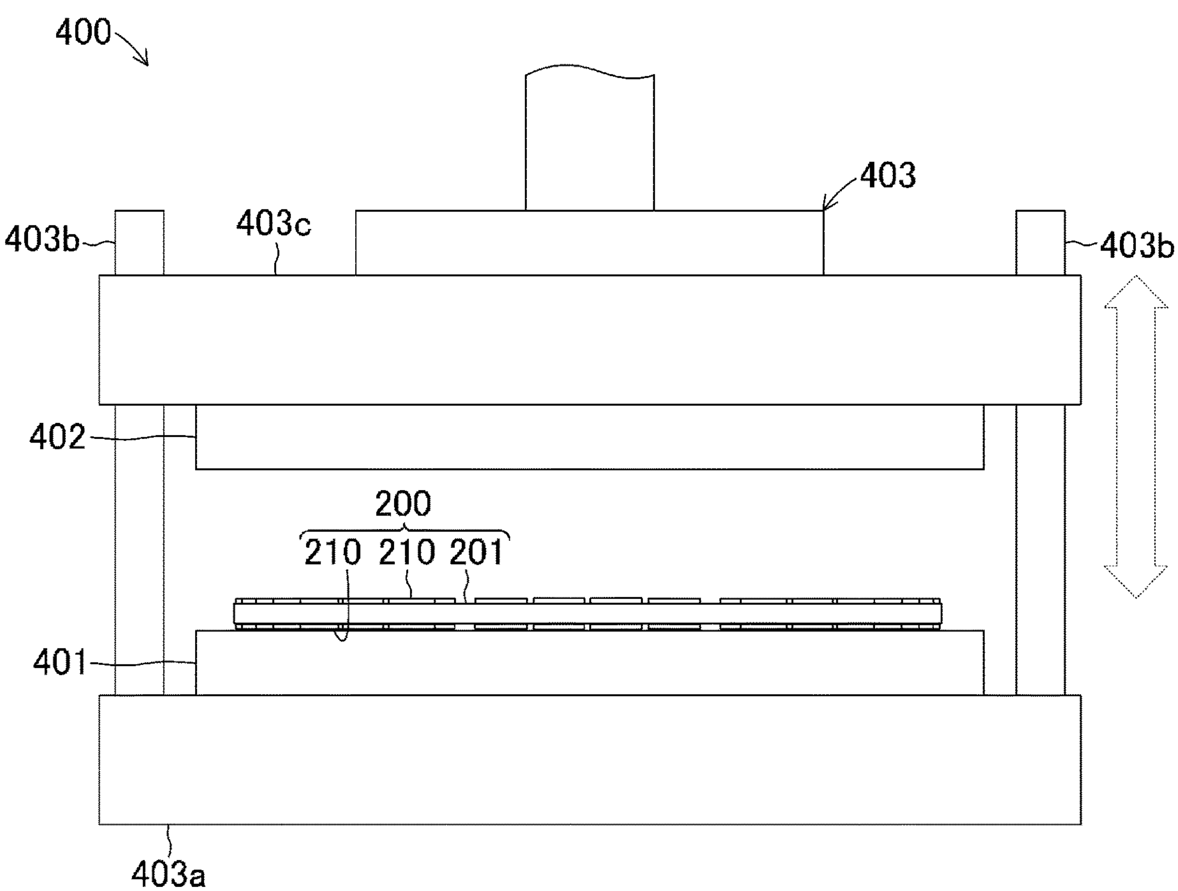

212
212a 212b
210

FIG. 8


400
403
403b
403c
403b
402
200
210 210 201
401
403a

METHOD FOR MANUFACTURING WET FRICTION PLATE

TECHNICAL FIELD

The present invention relates to a method for producing a wet friction plate used in lubricating oil.

BACKGROUND ART

A wet multiple disc clutch device has been mounted on vehicles such as four- or two-wheeled motor vehicles in order to transmit or interrupt rotary driving force of a motor such as an engine to a driven body such as a wheel. In the wet multiple disc clutch device, two plates facing each other in lubricating oil are generally pushed against each other so that rotary driving force is transmitted or interrupted.

Of these two plates, one is configured with a flat-plate ring-shaped wet friction plate which includes friction materials disposed along the circumferential direction on the surface of a core metal. For example, Patent Literature 1 described below discloses a laser processing method and a laser processing device of a friction plate (hereinafter, referred to as a "wet friction plate"), which form a concave portion (hereinafter, referred to as a "fine groove") including a minute uneven portion or groove on the surface of a friction plate by laser light. This allows the wet friction plate to have a processed end in the fine groove sharpened by laser light. This can promote the circulation of lubricating oil in the fine groove, compared to a known cutting process.

CITATION LIST

Patent Literature

Patent Literature 1: JP-A-2014-133242

In the laser processing method and the laser processing device of a wet friction plate described in Patent Literature 1, the circulation of lubricating oil is promoted. On the other hand, the retention property of lubricating oil is low. Therefore, a problem is raised in that the cooling performance of a wet friction plate by lubricating oil deteriorates.

The present invention addresses the above-described problem. An object of the present invention is to provide a method for producing a wet friction plate that can improve the retention property of lubricating oil in the fine groove formed by laser light.

SUMMARY OF INVENTION

In order to achieve the above object, as a characteristic of the present invention, there is provided a method for producing a wet friction plate that includes a friction material disposed along a circumferential direction on a surface of a core metal formed in a flat-plate ring shape. The method for producing a wet friction plate includes: a fine groove forming step of irradiating the friction material with laser light that is displaced relative to the friction material, thereby to form a concave fine groove on a surface of the friction material; and a crushing step of pressing the friction material to crush and deform the fine groove.

According to the characteristic of the present invention configured in this manner, in the method for producing a wet friction plate, the fine groove is formed by laser light. Thereafter, the friction material having the formed fine groove is pressed. Accordingly, the fine groove is crushed. Therefore, the surface inside the fine groove is formed into

an uneven shape. This can improve the retention property of lubricating oil. In this case, the fine groove has a portion which has a groove width of 10 μm or more and 1000 μm or less and a depth of 10 μm or more and 1000 μm or less and which is dug such that the cross-sectional shape of the fine groove is a concave shape. This portion intermittently or continuously extends in a long-length manner. In this case, a rubber material or a cork material as well as a paper material that is an aggregate of fibers can be used as the friction material.

Moreover, as another characteristic of the present invention, the method for producing a wet friction plate further includes a friction material disposing step of disposing the friction material on the core metal, and the fine groove forming step is performed before the friction material disposing step.

According to the another characteristic of the present invention configured in this manner, in the method for producing a wet friction plate, the fine groove forming step is performed before the friction material disposing step. Therefore, the fine groove can be simply and precisely formed, compared to when the processing for the fine groove is performed to the friction material on the core metal.

Moreover, as another characteristic of the present invention, in the method for producing a wet friction plate, the crushing step is performed to the friction material disposed on the core metal.

According to the another characteristic of the present invention configured in this manner, in the method for producing a wet friction plate, the crushing step is performed to the friction material disposed on the metal core. Therefore, compared to when the crushing step is performed to a location other than the core metal, the crushing step can also serve as a step of sticking the friction material on the core metal. This can reduce the man-hours, which enables efficient production of the wet friction plate.

Moreover, as another characteristic of the present invention, the method for producing a wet friction plate further includes a friction material disposing step of disposing the friction material on the core metal, the fine groove forming step is performed to the friction material disposed on the core metal, and the crushing step is performed to the friction material disposed on the core metal.

According to the another characteristic of the present invention configured in this manner, in the method for producing a wet friction plate, the fine groove forming step and the crushing step are performed to the friction material disposed on the metal core. This enables the fine groove to be precisely formed in an accurate position on the core metal.

Moreover, as another characteristic of the present invention, the method for producing a wet friction plate further includes a friction material producing step of producing, as the friction material, a paper body that includes an aggregate of numerous fibers impregnated with thermosetting resin, and in the crushing step, the friction material is pressed while heated.

According to the another characteristic of the present invention configured in this manner, in the method for producing a wet friction plate, the friction material configured with a paper body that includes an aggregate of numerous fibers impregnated with thermosetting resin is pressed while heated. This restrains springback of the pressed friction material. Accordingly, the fine groove can be precisely crushed to produce a wet friction plate.

Moreover, as another characteristic of the present invention, in the method for producing a wet friction plate, the paper body containing the thermosetting resin in a semi-cured state is produced in the friction material producing step, and the friction material is heated so that the thermosetting resin in the semi-cured state is completely cured in the crushing step.

According to the another characteristic of the present invention configured in this manner, in the method for producing a wet friction plate, the friction material is heated in the crushing step. Accordingly, the thermosetting resin in the semi-cured state is completely cured. This restrains springback of the pressed friction material. Accordingly, the fine groove can be precisely crushed to produce a wet friction plate. Also, in this method for producing a wet friction plate, the thermosetting resin is in a semi-cured state. Therefore, a pressure with which the friction material is pressed can be reduced, compared to when a friction material containing completely cured thermosetting resin is pressed.

Moreover, as another characteristic of the present invention, in the method for producing a wet friction plate, an adhesive agent is disposed between on the core metal and on the friction material in the friction material disposing step, and the friction material is pressed so that the friction material is stuck on the core metal, in the crushing step.

According to the another characteristic of the present invention configured in this manner, in the method for producing a wet friction plate, an adhesive agent is disposed between the core metal and the friction material. Then, the friction material is pressed. Accordingly, the friction material is stuck on the core metal. Therefore, crushing of the fine groove and sticking of the friction material can be simultaneously performed. As a result, working efficiency can be improved.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a cross-sectional view illustrating an entire configuration of a wet multiple disc clutch device including a wet friction plate according to the present invention.

FIG. **2** is a plan view schematically illustrating an appearance configuration of a wet friction plate according to the present invention to be incorporated into the wet multiple disc clutch device illustrated in FIG. **1**.

FIG. **3** is a partially enlarged cross-sectional view illustrating a configuration of a fine groove formed on a friction material in the wet friction plate illustrated in FIG. **2**.

FIG. **4** is a flowchart illustrating steps for producing a wet friction plate according to the present invention.

FIG. **5** is a side view schematically illustrating a configuration of a laser processor used in a fine groove forming step in the steps of producing a wet friction plate illustrated in FIG. **4**.

FIG. **6** is a plan view illustrating a long-length friction material having fine grooves formed by the laser processor illustrated in FIG. **5**.

FIG. **7** is a partially enlarged cross-sectional view illustrating a configuration of the fine groove by the laser processor illustrated in FIG. **5**.

FIG. **8** is a side view schematically illustrating a configuration of a hot press processor used in a fine groove crushing step in the steps of producing a wet friction plate illustrated in FIG. **4**.

DESCRIPTION OF EMBODIMENTS

Hereinafter, one embodiment of the method for producing a wet friction plate according to the present invention will be described with reference to the drawings. FIG. **1** is a cross-sectional view schematically illustrating an entire configuration of a wet multiple disc clutch device **100** including a wet friction plate **200** according to the present invention. Also, FIG. **2** is a plan view schematically illustrating an appearance configuration of the wet friction plate **200** according to the present invention included by the wet multiple disc clutch device **100** illustrated in FIG. **1**. It is noted that in the drawings referred herein, constituents are partly depicted schematically, for example, in an exaggerated manner, for facilitating understanding of the present invention. Therefore, the dimension, ratio, and others of each constituent may be varied.

This wet multiple disc clutch device **100** is a mechanical device for transmitting or interrupting driving force of an engine (not illustrated) as a motor in a two-wheeled motor vehicle (motorcycle) to a wheel (not illustrated) as a driven body. The wet multiple disc clutch device **100** is disposed between the engine and a transmission (not illustrated).

(Configuration of Wet Multiple Disc Clutch Device **100**)

The wet multiple disc clutch device **100** includes a housing **101** made of aluminum alloy. The housing **101** is a member that configures part of the casing of the wet multiple disc clutch device **100**. The housing **101** has a bottomed cylindrical shape. A side surface on the illustrated left side of this housing **101** is fixed with an input gear **102** through a torque damper **102***a* by a rivet **102***b*. The input gear **102** meshes with an unillustrated drive gear that is driven to rotate by a drive of the engine, and is accordingly driven to rotate. On the inner circumferential surface of the housing **101**, a plurality of (eight, in the present embodiment) clutch plates **103** are retained by spline engagement in a state in which they can be displaced along the axial direction of the housing **101** and rotated integrally with the housing **101**.

The clutch plates **103** are each a flat-plate ring-shaped component to be pushed against the wet friction plate **200** described later. The clutch plate **103** is obtained by punching a ring shape from a thin plate material formed of steel plate cold commercial (SPCC). Unillustrated oil grooves described later are formed on both surfaces (front and back surfaces) of the clutch plate **103**. The oil grooves each have a depth of several μm to several tens μm such that lubricating oil is retained. Also, both side surfaces (front and back surfaces) having oil grooves formed thereon of the clutch plate **103** are subjected to a surface curing treatment for a purpose of improving abrasion resistance. It is noted that this surface curing treatment is not directly involved in the present invention. Therefore, explanation of this surface curing treatment will be omitted.

Inside the housing **101**, a friction plate holder **104** formed in a substantially cylindrical shape is disposed coaxially with the housing **101**. On the inner circumferential surface of this friction plate holder **104**, a large number of spline grooves is formed along the axial direction of the friction plate holder **104**. Into the spline grooves, a shaft **105** is spline-fitted. The shaft **105** is a hollow axial body. One end side (illustrated right side) of the axial body rotatably supports the input gear **102** and the housing **101** via a needle bearing **105***a*. At the same time, the end side fixes and supports, via a nut **105***b*, the friction plate holder **104** into which the end side is spline-fitted. That is, the friction plate holder **104** rotates integrally with the shaft **105**. On the other hand, the other end (illustrated left side) of the shaft **105** is linked to the unillustrated transmission of a two-wheeled motor vehicle.

An axial push rod **106** extends through the hollow part of the shaft **105** and projects from the one end (illustrated right side) of the shaft **105**. The push rod **106** is linked to an unillustrated clutch operation lever of a two-wheeled motor vehicle at a side (illustrated left side) opposite to an end projecting from one end (illustrated right side) of the shaft **105**. The push rod **106** slides along the axial direction of the shaft **105** inside the hollow part of the shaft **105** in response to operation of the clutch operation lever.

A plurality of (seven, in the present embodiment) wet friction plates **200** are retained on the outer circumferential surface of the friction plate holder **104** by spline fitting in a state of sandwiching the clutch plates **103**, in such a manner that the wet friction plates **200** can be displaced along the axial direction of the friction plate holder **104** and rotated integrally with the friction plate holder **104**.

On the other hand, the inside of the friction plate holder **104** is filled with a prescribed amount of lubricating oil (not illustrated). At the same time, three tubular support columns **104***a* are formed (only one is illustrated). Lubricating oil is supplied between the wet friction plate **200** and the clutch plate **103** to absorb friction heat generated between the wet friction plate **200** and the clutch plate **103** and prevent wear of friction material **210**.

Also, the three tubular support columns **104***a* each project toward the outside (illustrated right side) in the axial direction of the friction plate holder **104**. A pressing cover **107** disposed coaxially with the friction plate holder **104** is assembled to each of the tubular support columns **104***a* via a bolt **108***a*, a backing plate **108***b*, and a coil spring **108***c*. The pressing cover **107** has a substantial circular plate shape with an outer diameter that is substantially the same as the outer diameter of the wet friction plate **200**. The pressing cover **107** is pressed to the friction plate holder **104** side by the coil spring **108***c*. Also, a release bearing **107***a* is disposed on the inner center of the pressing cover **107** in a position facing the tip end portion on the illustrated right side of the push rod **106**.

(Configuration of Wet Friction Plate **200**)

In particular, as illustrated in FIG. **2**, the wet friction plate **200** includes an oil groove **203** and a friction material **210** on a flat-plate ring-shaped core metal **201**. The core metal **201** is a member that serves as a base of the wet friction plate **200**. The core metal **201** is obtained by punching a ring shape from a thin plate material formed of steel plate cold commercial (SPCC). In this case, an internally-toothed spline **202** for spline fitting with the friction plate holder **104** is formed on the inner circumferential portion of the core metal **201**.

On a side facing the clutch plate **103** of this wet friction plate **200**, that is, on a ring-shaped plate surface facing the clutch plate **103** of the core metal **201**, a plurality of (32, in the present embodiment) small piece-shaped friction materials **210** is disposed along the circumferential direction of the core metal **201** with the oil grooves **203**, each configured as a space, interposed between the friction materials **210**.

The oil grooves **203** are each a channel that guides lubricating oil between the inner circumferential edge and the outer circumferential edge of the core metal **201** of the wet friction plate **200**. Also, the oil groove **203** is an oil retaining portion for allowing lubricating oil to exist between the wet friction plate **200** and the clutch plate **103**. The oil groove **203** is configured as a space between the friction material **210** and the friction material **210** which are next to each other. In the present embodiment, the oil groove **203** includes a fan-shaped portion and a linearly extending portion. The fan-shaped portion is formed between every four small piece-shaped friction materials **210**. The linearly extending portion is formed each between the four friction materials **210** disposed between two fan-shaped oil grooves **203**. It is noted that the geometry and number of the oil grooves **203** are appropriately set depending on the specification of the wet friction plate **200**.

The friction materials **210** increase the frictional force on the clutch plate **103**. The friction materials **210** are configured with small piece-shaped paper materials that are stuck along the circumferential direction of the core metal **201**. More particularly, the friction material **210** is configured with a paper body impregnated with a thermosetting resin subsequently cured.

Here, the paper body is configured with a filler which is added to an aggregate of at least one of organic fibers and inorganic fibers. Here, the organic fibers to configure the paper body may be one or a combination of two or more of wood pulp, synthetic pulp, polyester-based fiber, polyamide-based fiber, polyimide-based fiber, polyvinyl alcohol modified fiber, polyvinyl chloride fiber, polypropylene fiber, polybenzimidazole fiber, acryl fiber, carbon fiber, phenol fiber, nylon fiber, cellulose fiber, and others. The inorganic fibers to configure the paper body may be one or a combination of two or more of glass fiber, rock wool, potassium titanate fiber, ceramic fiber, silica fiber, silica-alumina fiber, kaolin fiber, bauxite fiber, kayanoid fiber, boron fiber, magnesia fiber, metal fiber, and others.

The filler exerts a function as a friction adjuster and/or a solid lubricant. The filler may be one or a combination of two or more of barium sulfate, calcium carbonate, magnesium carbonate, silicon carbide, boron carbide, titanium carbide, silicon nitride, boron nitride, alumina, silica, zirconia, cashew dust, rubber dust, diatomaceous earth, graphite, talc, kaolin, magnesium oxide, molybdenum disulfide, nitrile rubber, acrylonitrile-butadiene rubber, styrene butadiene rubber, silicon rubber, fluorine rubber, and others. Also, examples of thermosetting resin include phenol-based resin, melamine resin, epoxy resin, urea resin, and silicone resin.

This friction material **210** has a thickness of 0.3 mm or more and 0.6 mm or less. The friction material **210** is stuck on the core metal **201** through an unillustrated adhesive agent. In the present embodiment, the friction materials **210** are configured with small-piece groups each including four small pieces that have a quadrangular shape extending in the circumferential direction of the core metal **201** and are disposed in the circumferential direction of the core metal **201** with three linear oil grooves **203** interposed between the friction materials **210**. At the same time, the friction materials **210** are configured with eight small-piece groups described above disposed in the circumferential direction of the core metal **201** with eight fan-shaped oil grooves **203** interposed between the eight small-piece groups. It is noted that the geometry and number of the friction materials **210** are not limited to the present embodiment. The geometry and number of the friction materials **210** are appropriately set depending on the specification of the wet friction plate **200**.

A fine groove **211** is formed on the surface of this friction material **210**. The fine groove **211** is a portion for defining the retention property and the discharge property of lubricating oil on the friction material **210**. The fine groove **211** has a groove shape which is recessed in a concave manner on the surface of the friction material **210**. More specifically, as illustrated in FIG. **3**, the fine groove **211** has a groove shape that opens on the surface of the friction material **210** and is inwardly recessed in a concave manner. In this case, an intra-groove surface **212** that forms the fine groove **211** is configured with an uneven surface having an irregular uneven portion.

In the present embodiment, the fine groove **211** has an inclined surface in which two side surfaces **212***a* and **212***b* approach each other toward the inside of the friction material **210**. That is, the cross-sectional shape of the fine groove **211** is a substantial V shape (triangular shape). In this case, the fine groove **211** is formed such that it has a groove width of about 100 μm and a deepest depth of about 200 μm in the present embodiment.

This fine groove **211** has an arc shape which continuously extends along the circumferential direction of the core metal **201** on the friction material **210**. In the present embodiment, the fine groove **211** is formed by three arcs aligning in the radial direction of the wet friction plate **200**. In this case, the three arcs are each configured with a circle having the same center as the center of the core metal **201**.

(Production of Wet Friction Plate **200**)

Next, a method for producing the wet friction plate **200** configured in this manner will be described with reference to FIG. **4**. First, an operator produces the friction material **210** as a first step (friction material producing step). Specifically, the producing step of the friction material **210** mainly includes a sheet making step and a curing step.

The sheet making step is a step of filtering out fibers dispersed in liquid and forming a long-length sheet-shaped paper body. The sheet making step is a known method. Specifically, the sheet making step is a work of drying a raw material filtered out from a raw material liquid in a slurry state into a long-length sheet-like shape to obtain a long-length sheet-shaped paper body. The raw material liquid is prepared by stirring a raw material of a paper body poured in water, i.e., the above-described organic fibers and/or the above-described inorganic fibers, a filler, and an aggregating agent. In this case, the paper body is dried until a moisture content of 10% or less is reached.

Next, the curing step is a work of curing thermosetting resin with which the paper body is impregnated. Specifically, the operator sprays thermosetting resin liquid (phenol-based resin liquid) on the paper body or immerses the paper body in thermosetting resin liquid, so that the paper body is impregnated with the thermosetting resin liquid. Thereafter, the thermosetting resin liquid is cured through heating by a heater, a heated roller, or the like. In this case, the operator cures the thermosetting resin liquid into a semi-cured state. In the semi-cured state, curing of the thermosetting resin liquid has proceeded in a range in which the thermosetting resin liquid changing into solid is not completely cured. Accordingly, the operator can obtain the friction material **210** configured with the paper body containing the thermosetting resin liquid in the semi-cured state.

Next, the operator forms the fine groove **211** on the friction material **210** as a second step (fine groove forming step). In this case, the operator uses a laser processor **300** to form the fine groove **211**. Here, the laser processor **300** is a mechanical device for irradiating the friction material **210** with laser light L to form the fine groove **211**.

As illustrated in FIG. **5**, this laser processor **300** mainly includes a laser oscillator (not illustrated), a laser adjustment optical system (not illustrated), a laser head **301**, a work table **302**, a work conveying mechanism **303**, and a controller (not illustrated). The laser oscillator is a mechanical device for emitting laser light L for forming the fine groove **211** on the friction material **210**. In the present embodiment, the laser oscillator is configured as a 60 W output oscillator for generating pulse laser light having a frequency of 300 kHz and a short pulse width of nanoseconds, picoseconds, femtoseconds, or the like. The laser adjustment optical system is configured with various optical elements such as a lens and a mirror and an optical component including optical fibers and others. The various optical elements guide laser light L to the laser head **301** while performing various adjustments such as correction of the beam diameter, beam shape, and aberration of laser light L emitted from the laser oscillator.

The laser head **301** is an optical device that emits laser light L guided from the laser adjustment optical system toward the work table **302** to collect light onto the wet friction plate. This laser head **301** is configured such that it can be displaced relative to the work table **302** in three axis directions of X-axis direction, Y-axis direction, and Z-axis direction which are orthogonal to one another. It is noted that the laser oscillator, the laser adjustment optical system, and the laser head **301** themselves are known mechanical devices.

The work table **302** is a board that supports the friction material **210** from below in a position facing the laser head **301**. The work table **302** is configured with a metal material formed in a flat plate shape. The work conveying mechanism **303** is a mechanical device for conveying the long-length friction material **210** from one side to the other side in the lengthwise direction. The work conveying mechanism **303** mainly includes a pair of drive rollers which sandwich the friction material **210**.

The controller is configured with a microcomputer including a CPU, a ROM, a RAM, and others. The controller comprehensively controls an entire action of the laser processor **300**. Specifically, the controller controls actions of the laser oscillator, the laser adjustment optical system, the laser head **301**, and the work conveying mechanism **303**, in response to instructions by the operator. In this manner, the controller disposes the friction material **210** on the work table **302**. At the same time, the controller displaces the laser head **301** while irradiating the friction material **210** positioned on the work table **302** with laser light L, such that the fine groove **211** is formed.

In this second step, the operator draws out the friction material **210** wound in a roll shape. At the same time, the end portion of the drawn friction material **210** is grasped by the work conveying mechanism **303**. Thereafter, the operator instructs the controller of the laser processor **300** to form the fine groove **211**. In response to this instruction, the controller controls the action of the work conveying mechanism **303** to intermittently convey the friction material **210**. This allows the friction material **210** to be intermittently positioned on the work table **302**. Subsequently, the controller displaces the laser head **301** in the X-axis direction and the Y-axis direction relative to the friction material **210** positioned on the work table **302** while allowing laser light L to be emitted from the laser head **301**. Accordingly, laser light L is displaced on the friction material **210** thereby to form the fine groove **211**.

In the present embodiment, the laser processor **300** forms, as illustrated in FIG. **6**, three fine grooves **211** formed on each of the above-described four friction materials **210** disposed between two fan-shaped oil grooves **203** described above, in such a state that the fine grooves **211** are continuously connected along the width direction of the friction material **210**. In this case, the formed fine grooves **211** each have, as illustrated in FIG. **7**, a V-shaped (triangular) cross-sectional shape. The side surfaces **212***a* and **212***b* which defines each of the fine grooves **211** have few uneven portions and are a substantially flat planar surface. Also, the fine grooves **211** are each formed so as to be deeper than the depth of the groove of the finally formed fine groove **211**. It is noted that in FIG. **6**, the friction material **210** in the fine groove **211** is virtually denoted by a double dot-and-dash line. Also, in FIG. **5** and FIG. **6**, the conveying direction of the long-length friction material **210** is denoted by a broken line arrow.

Next, the operator prepares the core metal **201** and disposes the friction material **210** on each of two plate surfaces of this core metal **201**, as a third step (friction material disposing step). Here, the core metal **201** is formed in an annular shape having the splines **202** by separate press processing. This press processing is a known method. Therefore, explanation of the press processing will be omitted.

The operator applies an adhesive agent in a liquid state on the entire surface of the plate surface of the core metal **201** using a tool such as a brush or a roller. Thereafter, the friction material **210** is placed on this adhesive agent. Here, thermosetting resin in a liquid state is used as the adhesive agent. Also, the operator may cut the friction material **210** extending in a band shape in a state of being placed on the core metal **201**, to form the small piece-shaped friction material **210**. Alternatively, the friction materials **210**, which are obtained by previously cutting the friction material **210** extending in a band shape into small pieces, may be placed on the core metal **201**. A method for disposing these friction materials **210** on the core metal **201** is a known technique.

The friction materials **210** disposed on the core metal **201** are in a state of being temporarily fixed on the core metal **201** by adhesive force of the uncured adhesive agent. Therefore, the operator can dispose the friction materials **210** on both surfaces by turning the core metal **201** upside down. It is noted that in this third step, the operator can apply an adhesive agent only on a position where the friction materials **210** are to be disposed. Alternatively, the operator can apply an adhesive agent only in an annular shape along the circumferential direction where the friction materials **210** are to be placed. Also, the operator can apply an adhesive agent on the friction materials **210**.

Next, the operator performs, as a fourth step, crushing processing to the fine groove **211** (crushing step). This crushing processing also serves as sticking processing of the friction material **210** on the core metal **201**. The crushing step is performed using a hot press processor **400**. Here, the hot press processor **400** is a mechanical device for pressing the friction material **210** while heating, to crush the fine groove **211** and fix the friction material **210** to the core metal **201**. This hot press processor **400** is a known mechanical device. Therefore, although the configuration of the hot press processor **400** will be briefly described below, detailed explanation thereof will be omitted.

This hot press processor **400** mainly includes, as illustrated in FIG. **8**, a lower pressing plate **401**, an upper pressing plate **402**, a movable support device **403**, and a controller. The lower pressing plate **401** is a component on which the core metal **201** is to be placed. At the same time, the lower pressing plate **401** operates together with the upper pressing plate **402** to sandwich the core metal **201** for heating and pressing. The lower pressing plate **401** is configured with a metal material formed in a plate shape. In this case, a heater (not illustrated) to perform heating by energization is disposed inside the lower pressing plate **401**. This lower pressing plate **401** is fixed to the upper surface of a lower base **403***a* of the movable support device **403**.

The upper pressing plate **402** is a component that is disposed to face the top of the lower pressing plate **401**. The upper pressing plate **402** operates together with the lower pressing plate **401** to sandwich the core metal **201** for heating and pressing. The upper pressing plate **402** is configured with a metal material formed in a plate shape. A heater (not illustrated) to perform heating by energization is disposed inside this upper pressing plate **402**. This upper pressing plate **402** is fixed to a lower surface of an upper base **403***c* of the movable support device **403**.

The movable support device **403** is a mechanical device that supports the upper pressing plate **402** such that it can approach or move away from the lower pressing plate **401** (see the illustrated broken line arrow). The movable support device **403** includes the lower base **403***a*, a support column **403***b*, and the upper base **403***c*. The lower base **403***a* is a metal flat plate body that fixes and supports the lower pressing plate **401**. The support column **403***b* is a component that supports the upper base **403***c* while guiding it in a direction in which the upper pressing plate **402** approaches or moves away from the lower pressing plate **401**. The support column **403***b* is configured with a plurality of metal bar bodies rising at the outer edge of the lower base **403***a*.

The upper base **403***c* is a metal flat plate body that supports the upper pressing plate **402**. Through holes are formed in a position corresponding to the support column **403***b* at the outer edge of this upper base **403***c*. The support column **403***b* slidably fits into each of these through holes. Also, this upper base **403***c* is supported by a hydraulic driving device (not illustrated) that raises or lowers the upper base **403***c* by hydraulic pressure.

The controller is configured with a microcomputer including a CPU, a ROM, a RAM, and others. The controller comprehensively controls an entire action of the hot press processor **400**. Specifically, the controller controls actions of heaters in the lower pressing plate **401** and the upper pressing plate **402** and the hydraulic driving device in the movable support device **403** in response to instructions by the operator. In this manner, the core metal **201** is pressed by the lower pressing plate **401** and the upper pressing plate **402** under heating. Accordingly, the friction material **210** including the fine groove **211** is crushed.

In this fourth step, the operator places, onto the lower pressing plate **401**, the core metal **201** having the temporarily fixed friction material **210**. Thereafter, the operator instructs the controller of the hot press processor **400** to heat and press the friction material **210**. In response to this instruction, the controller controls actions of heaters in the lower pressing plate **401** and the upper pressing plate **402**. In this manner, the lower pressing plate **401** and the upper pressing plate **402** are heated to a prescribed temperature. Thereafter, the controller controls an action of the hydraulic driving device to lower the upper pressing plate **402** toward the lower pressing plate **401** side. In this manner, the entirety of the core metal **201** is pressed (see the illustrated broken line arrow).

Accordingly, the core metal **201** is heated and pressed in a state of being sandwiched between the lower pressing plate **401** and the upper pressing plate **402**. In this case, a temperature for heating the friction material **210**, a pressure for pressurizing the friction material **210**, and a time taken for the heating and pressurizing are appropriately set depending on the specification of the wet friction plate **200**. However, at least a pressure and time for compressing the friction material **210** in the thickness direction is necessary. Furthermore, a temperature and time for completely solidifying thermosetting resin is necessary. Here, complete solidification of thermosetting resin indicates that thermosetting resin is solidified enough to endure use of the wet friction plate **200**.

Accordingly, in the core metal **201**, thermosetting resin in a semi-cured state contained inside the friction material **210** and thermosetting resin applied between the core metal **201** and the friction material **210** are both solidified. Therefore, the friction material **210** is solidified in a state of being compressed in the thickness direction. At the same time, the friction material **210** is fixed and stuck on the core metal **201**. In this case, the friction material **210** is fixed and stuck on the core metal **201** in a state of being compressed such that the thickness becomes 1/10 or more and 1/2 or less of the thickness before pressurization.

Also, the fine groove **211** formed on the friction material **210** is crushed in the thickness direction of the friction material **210**, as illustrated in FIG. **3**. Specifically, the depth of the groove of the fine groove **211** is reduced. At the same time, the pointed bottom shape of the groove is deformed into a broad shape. Also, unevenness occurs on the side surfaces **212***a* and **212***b*. Therefore, the cross-sectional shape becomes, as an entirety, a substantial V shape which is crushed in the thickness direction of the friction material **210**.

Further, the controller controls such that the core metal **201** is heated and pressed by the lower pressing plate **401** and the upper pressing plate **402** for a prescribed time. Thereafter, the controller controls an action of the hydraulic driving device to raise the upper pressing plate **402**, so that the upper pressing plate **402** moves away from the lower pressing plate **401**. At the same time, the controller stops heating by heaters of the lower pressing plate **401** and the upper pressing plate **402** (see the illustrated broken line arrow). Thus, the operator can take out the core metal **201**, on which the friction material **210** is fixed and stuck, i.e., the wet friction plate **200**, from on the lower pressing plate **401**. Thereafter, the operator performs an adjusting step and a test step of the friction property to complete the wet friction plate **200**. Since these steps are not directly involved in the present invention, explanation thereof will be omitted.

(Action of Wet Friction Plate **200**)

Next, an action of the wet friction plate **200** configured in this manner will be described. As described above, this wet friction plate **200** is assembled in the wet multiple disc clutch device **100** for use. Then, as described above, this wet multiple disc clutch device **100** is disposed between the engine and the transmission in the vehicle. In response to an operation of a clutch operation lever by an operator of the vehicle, driving force of the engine is transmitted to the transmission or interrupted.

That is, when the operator of the vehicle (not illustrated) operates a clutch operation lever (not illustrated) to retreat (displace to the illustrated left side) the push rod **106**, the tip end of the push rod **106** is in a state of not pressing the release bearing **107***a*. Therefore, the pressing cover **107** presses the clutch plate **103** by elastic force of the coil spring **108***c*. Accordingly, the clutch plate **103** and the wet friction plate **200** are in a state of being friction connected. In this state, the clutch plate **103** and the wet friction plate **200** are pushed against each other while being displaced toward a side of a receiving part **104***b* formed in a flange shape on the outer circumferential surface of the friction plate holder **104**. As a result, driving force of the engine, transmitted to the input gear **102**, is transmitted to the transmission via the clutch plate **103**, the wet friction plate **200**, the friction plate holder **104**, and the shaft **105**.

On the other hand, when the operator of the vehicle operates the clutch operation lever (not illustrated) to advance (displace to the illustrated right side) the push rod **106**, the tip end of the push rod **106** is in a state of pressing the release bearing **107***a*. Accordingly, the pressing cover **107** is displaced to the illustrated right side against elastic force of the coil spring **108***c*. Then, the pressing cover **107** and the clutch plate **103** move away from each other. This displaces the clutch plate **103** and the wet friction plate **200** to the pressing cover **107** side and releases the state in which they are pushed against each other to be connected. In this manner, the clutch plate **103** and the wet friction plate **200** move away from each other. As a result, driving force from the clutch plate **103** to the wet friction plate **200** comes not to be transmitted. In this manner, transfer of driving force of the engine transmitted to the input gear **102** to the transmission is interrupted.

In a clutch ON state, this clutch plate **103** and the wet friction plate **200** are in friction contact with each other. In this state, lubricating oil existing on the friction material **210** partly enters the fine groove **211**, in the fine groove **211** formed on the surface layer of the friction material **210**. In this case, the surface area of the intra-groove surface **212** of the fine groove **211** increases because of unevenness. This increases the amount of lubricating oil penetrating into the internal structure of the friction material **210**. As a result, the cooling effect of the friction material **210** can be improved.

In a clutch OFF state, the clutch plate **103** and the wet friction plate **200** are spaced apart from each other. In this state, the retention property of lubricating oil in the fine groove **211** is improved by unevenness formed on the intra-groove surface **212**, in the fine groove **211**. Therefore, the cooling effect of the friction material **210** can be improved.

Also, even when the clutch plate **103** and the wet friction plate **200** are pressure contacted and separated in a repeated manner, the bottom formed broadly in a non-pointed manner of the fine groove **211** can restrain occurrence or progress of breaks or cracks of the fine groove **211**. In this manner, durability of the friction material **210** can be improved.

As understood from the above-described explanation of actions, according to the above-described embodiment, the fine groove **211** is crushed to form the side surfaces **212***a* and **212***b* having a heavily contoured uneven surface, even when the fine groove **211** is formed by laser, in the method for producing the wet friction plate **200**. Therefore, the retention property of lubricating oil in the fine groove **211** can be improved.

Furthermore, embodiments of the present invention are not limited to the above-described embodiment. Various modifications are possible as long as such modifications do not depart from the object of the present invention. It is noted that in modification examples described below, a constituent part similar to the wet friction plate **200** in the above-described embodiment is assigned with a reference sign corresponding to the reference sign assigned to the wet friction plate **200**. Also, explanation thereof will be omitted.

For example, in the above-described embodiment, the cross-sectional shape of the fine groove **211** is a substantial V shape (triangular shape). However, the intra-groove surface **212** is formed in a distorted shape by the crushing step. Therefore, the fine groove **211** can also be formed into a shape other than the V shape, such as a U shape or an arc shape. Also, the fine groove **211** can be formed into a shape other than these shapes.

Also, in the above-described embodiment, the fine groove **211** has a groove width of about 100 μm and a deepest depth of about 200 μm. However, the fine groove **211** ay have a width of 10 μm or more and 1000 μm or less. Also, the fine groove **211** may have a depth of 10 μm or more and 1000 μm or less.

Also, in the above-described embodiment, the fine groove **211** has an arc shape which continuously extends along the circumferential direction of the core metal **201** on the friction material **210**. However, the fine groove **211** may extend in the radial direction of the core metal **201** or other directions. In this case, the fine groove **211** may also extend in the circumferential direction of the core metal **201**, the radial direction of the core metal **201**, or other directions, in a continuous linear or curved manner. Alternatively, the fine groove **211** may be formed in an intermittent linear or curved manner.

Also, in the above-described embodiment, the fine groove **211** is formed by three arcs aligning in the radial direction of the wet friction plate **200**. In this case, the three arcs are each configured with a circle having the same center as the center of the core metal **201**. However, at least one fine groove **211** may be formed. Also, when the fine groove **211** is configured with a plurality of arcs aligning in the radial direction of the wet friction plate **200**, the arcs may naturally have radiuses different from one another.

Also, in the above-described embodiment, the fine groove **211** having a substantial V-shaped (triangular) cross-sectional shape is formed in the step of forming the fine groove **211** in the second step. However, in the step of forming the fine groove **211** in the second step, a shape other than a V shape, such as a U shape, a rectangular shape, or an arc shape, may also be formed. In addition, the fine groove **211** can also be formed in a shape other than these.

Also, in the above-described embodiment, the step of forming the fine groove **211** was performed to the friction material **210** before disposed on the core metal **201**. Accordingly, the operator can simply form the highly precise fine groove **211**, compared to when the fine groove **211** is formed to the friction material on the core metal. In this case, the operator can reliably prevent the fine groove **211** from being mistakenly formed on the core metal **201**. However, naturally, the step of forming the fine groove **211** may be performed to the friction material **210** disposed on the core metal **201**. This enables the operator to precisely form the fine groove **211** in an accurate position on the core metal **201**.

Also, in the above-described embodiment, the step of crushing the fine groove **211** is performed to the friction material **210** disposed on the core metal **201**. Accordingly, the operator can also perform the step of sticking the friction material **210** onto the core metal **201** at the same time. This can reduce the man-hours, which enables efficient production of the wet friction plate **200**. However, naturally, the step of crushing the fine groove **211** may be performed to the friction material **210** before disposed on the core metal **201**. This enables the operator to form the friction material **210** of different type or the fine groove **211** of different type by crushing in a separate step.

Also, in the above-described embodiment, the step of producing the friction material **210** is performed in the first step. However, in the method for producing a wet friction plate according to the present invention, a commercially available paper body or friction material **210** may be acquired instead of producing the friction material **210**.

Also, in the above-described embodiment, the thermosetting resin is cured into a semi-cured state in the step of producing the friction material **210** in the first step. Accordingly, the operator can restrain springback of the pressed friction material **210** and precisely crush the fine groove **211** in the step of crushing the fine groove **211** in the fourth step to produce the wet friction plate **200**. Also, in this case, in the step of crushing the fine groove **211** in the fourth step, a pressure with which the friction material **210** is pressed can be reduced, compared to when the friction material **210** containing completely cured thermosetting resin is pressed. However, in the step of producing the friction material **210** in the first step, thermosetting resin may also be completely cured. In this case, in the step of crushing the fine groove **211** in the fourth step, the friction material **210** containing completely cured thermosetting resin is pressed to crush the fine groove **211**.

Also, in the above-described embodiment, the friction material **210** is pressed while heated in the step of crushing the fine groove **211** in the fourth step. However, the friction material **210** may be merely pressed without being heated in the step of crushing the fine groove **211** in the fourth step. Also, in this method for producing the wet friction plate **200**, the friction material **210** may be pressed after the friction material **210** on the core metal **201** is stuck on the core metal **201** by heating. The step of crushing the fine groove **211** can also be executed in this manner.

Also, in the above-described embodiment, thermosetting resin is used as an adhesive agent for fixing the friction material **210** on the core metal **201**. However, naturally, an adhesive agent other than thermosetting resin may be used as the adhesive agent for fixing the friction material **210** on the core metal **201**. Examples of the adhesive agent include an elastomer-based adhesive agent (such as a silicone-based or modified silicone-based adhesive agent), a thermoplastic resin-based adhesive agent (such as a polyvinyl alcohol-based, polyamide-based, or polyolefin-based adhesive agent), and an inorganic adhesive agent (a ceramic-based adhesive agent or silicate of soda).

Also, in the above-described embodiment, the laser processor **300** was configured such that the laser head **301** can be displaced in three axis directions orthogonal to one another of X-axis direction, Y-axis direction, and Z-axis direction. However, the laser processor **300** may have a different configuration as long as the fine groove **211** can be formed to the friction material **210**. Therefore, for example, the laser processor **300** can be configured such that, instead of or in addition to displacing the laser head **301** and/or the work table **302**, a galvanometer scanner or a polygon mirror included in the laser head **301** scans laser light L in each of X-axis direction and Y-axis direction.

Also, in the above-described embodiment, the friction material **210** is configured with a paper body. However, the friction material **210** can also be configured with a material other than a paper material, such as a rubber material or a cork material.

Also, in the above-described embodiment, an example has been described, in which the wet friction plate according to the present invention is applied to the wet friction plate **200** used in the wet multiple disc clutch device **100**. However, the wet friction plate according to the present invention may be a wet friction plate used in oil. The wet friction plate according to the present invention can also be applied to, other than the wet multiple disc clutch device **100**, a wet friction plate used in a braking device for putting a brake on a rotation motion by a motor.

LIST OF REFERENCE SIGNS

L: Laser light, **100**: Wet multiple disc clutch device, **101**: Housing, **102**: Input gear, **102*a*:** Torque damper, **102*b*:** Rivet, **103**: Clutch plate, **104**: Friction plate holder, **104*a*:** Tubular support column, **104*b*:** Receiving part, **105**: Shaft, **105*a*:** Needle bearing, **105*b*:** Nut, **106**: Push rod, **107**: Pressing cover, **107*a*:** Release bearing, **108*a*:** Bolt, **108*b*:**

Receiving plate, **108***c:* Coil spring, **200**: Wet friction plate, **201**: Core metal, **202**: Spline, **203**: Oil groove, **210**: Friction material, **211**: Fine groove, **212**: Intra-groove surface, **212***a,* **212***b:* Side surface, **300**: Laser processor, **301**: Laser head, **302**: Work table, **303**: Work conveying mechanism, **400**: Hot press processor, **401**: Lower pressing plate, **402**: Upper pressing plate, **403**: Movable support device, **403***a:* Lower base, **403***b:* Support column, **403***c:* Upper base.

The invention claimed is:

1. A method for producing a wet friction plate that includes a friction material disposed along a circumferential direction on a surface of a core metal formed in a flat-plate ring shape,

the method comprising:

a friction material producing step of producing, as the friction material, a paper body that includes an aggregate of numerous fibers impregnated with thermosetting resin, the friction material producing step including curing the thermosetting resin until the thermosetting resin is not completely cured to produce the paper body containing the thermosetting resin in a semi-cured state;

a groove forming step of irradiating the friction material in which the paper body containing the thermosetting resin is in the semi-cured state with laser light that is displaced relative to the friction material, thereby to form a concave groove on a surface of the friction material; and

a crushing step of pressing the friction material while heated to crush and deform the groove formed by the laser light, wherein the friction material is heated so that the thermosetting resin in the semi-cured state is completely cured, and the groove is deformed so as to create unevenness on side surfaces of the groove to increase a surface area of an inner surface of the groove as compared to a surface area of the inner surface of the groove before the crushing step.

2. The method for producing a wet friction plate according to claim **1**, further comprising

a friction material disposing step of disposing the friction material on the core metal,

wherein the groove forming step is performed before the friction material disposing step.

3. The method for producing a wet friction plate according to claim **2**,

wherein the crushing step is performed to the friction material disposed on the core metal.

4. The method for producing a wet friction plate according to claim **3**, wherein

in the friction material disposing step, an adhesive agent is disposed between the core metal and the friction material, and

in the crushing step, the friction material is pressed so that the friction material is stuck on the core metal.

5. The method for producing a wet friction plate according to claim **1**, further comprising

a friction material disposing step of disposing the friction material on the core metal,

wherein the groove forming step is performed to the friction material disposed on the core metal, and

the crushing step is performed to the friction material disposed on the core metal.

6. A method for producing a wet friction plate that includes a friction material disposed along a circumferential direction on a surface of a core metal formed in a flat-plate ring shape,

the method comprising:

a friction material producing step of producing, as the friction material, a paper body impregnated with thermosetting resin, the paper body including an aggregate of numerous fibers to which a filler is added, the filler exerting a function as a friction adjuster and/or a solid lubricant, the friction material producing step including curing the thermosetting resin until the thermosetting resin is not completely cured to produce the paper body containing the thermosetting resin in a semi-cured state;

a groove forming step of irradiating the friction material in which the paper body containing the thermosetting resin is in the semi-cured state with laser light that is displaced relative to the friction material, thereby to form a concave groove on a surface of the friction material; and

a crushing step of pressing the friction material while heated to crush and deform the groove formed by the laser light, wherein the friction material is heated so that the thermosetting resin in the semi-cured state is completely cured, and the groove is deformed so as to create unevenness on side surfaces of the groove to increase a surface area of an inner surface of the groove as compared to a surface area of the inner surface of the groove before the crushing step.

7. A method for producing a wet friction plate that includes a friction material disposed along a circumferential direction on a surface of a core metal formed in a flat-plate ring shape,

the method comprising:

a friction material producing step of producing, as the friction material, a paper body impregnated with thermosetting resin, the paper body including an aggregate of numerous fibers to which a filler is added, the filler being one or more selected a group consisting of barium sulfate, calcium carbonate, magnesium carbonate, silicon carbide, boron carbide, titanium carbide, silicon nitride, boron nitride, alumina, silica, zirconia, cashew dust, rubber dust, diatomaceous earth, graphite, talc, kaolin, magnesium oxide, molybdenum disulfide, nitrile rubber, acrylonitrile-butadiene rubber, styrene butadiene rubber, silicon rubber and fluorine rubber, the friction material producing step including curing the thermosetting resin until the thermosetting resin is not completely cured to produce the paper body containing the thermosetting resin in a semi-cured state;

a groove forming step of irradiating the friction material in which the paper body containing the thermosetting resin is in the semi-cured state with laser light that is displaced relative to the friction material, thereby to form a concave groove on a surface of the friction material; and

a crushing step of pressing the friction material while heated to crush and deform the groove formed by the laser light, wherein the friction material is heated so that the thermosetting resin in the semi-cured state is completely cured, and the groove is deformed so as to create unevenness on side surfaces of the groove to increase a surface area of an inner surface of the groove as compared to a surface area of the inner surface of the groove before the crushing step.

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